

Clustering Taylor Swift’s Lyrics

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Motivation
Clustering of lyrics was suggested as a further project in the original poster.

Cluster Analysis uses numeric data to fit clusters, but text is non-numeric, so a map from words to numbers is necessary.

tf-idf
Term frequency – inverse document frequency (tf-idf) measures frequency of a term in a document.

Term frequency (tf) is the relative frequency of a term within a document:

$$tf(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

where t is the term and d is the document.

Inverse document frequency (idf) measures amount of information in the term:

$$idf(t, D) = \log \left(\frac{1 + N}{1 + n_t} \right) + 1$$

where D is the corpus, $N = |D|$, and n_t is the number of documents containing the term.

Then tf-idf is the product of these metrics:

$$tfidf(t, d, D) = tf(t, d) \cdot idf(t, D)$$

which has image

$$\left[0, \log \left(\frac{1 + N}{2} \right) + 1 \right]$$

Documents are represented by a vector, with each element being the tf-idf of a different word.

The Euclidean norm is used to scale vectors.

Sentence Transformers
Sentence Transformers are pre-trained models that map sentences to vector embeddings [2].

The all-mpnet-base-v2 model [3] maps sentences to a 768-dimensional vector space, preserving semantics.

The data can be clustered in this vector space or dimensionality reduction techniques can be applied.

Simulation
In an interview, Swift outlined three styles of writing: glitter gel pen, quill and fountain pen [4].

Knowing that there are three target clusters negates the need for intrinsic metric optimisation.

Two pipelines are used:
tf-idf Vectorisation -> K-Means and
Sentence Transformer -> K-Means.

Dataset
Select albums were chosen from a lyrics dataset obtained from Kaggle [5].

Dissimilarity Function
A dissimilarity function $d : M \times M \rightarrow R$ must follow three axioms:

- $d(x, y) \geq 0 \ \forall x, y \in M$
- $d(x, x) = 0 \ \forall x \in M$
- $d(x, y) = d(y, x) \ \forall x, y \in M$

Note that all distance functions are valid dissimilarity functions.

In text clustering, there are two main options for this function:

- L2 (Euclidean) – the straight line distance between two points.
- Cosine – a measure of the angle between two vectors.

Cosine is usually preferred - similar words such as 'like' and 'love' are close by this metric after applying a Sentence Transformer.

However, Swift’s quill writing style is classified by deliberate word choices [4], so L2 is more appropriate.

Comparison
Using tf-idf and K-Means, the contextual description of the resulting clusters are unclear.

song	Sentence Transformer	tf-idf
Gorgeous	glitter gel	2
peace	quill	2
Labyrinth	fountain	2

Summary
Sentence Transformers can capture more text information than tf-idf vectorisation methods.

Sentence Transformer (all-mpnet-base-v2) with K-Means shows the evolution in Swift’s writing and can find the three lyric-writing styles.

Future analysis could explore ensemble clustering to improve accuracy.

This clustering could also be used alongside musical data to classify Swift’s songs.

Code
Clustering and visualisations were done in Python, using numpy, pandas, scikit-learn and matplotlib.

Link: [github.com/lewis-tang-uk/Cluster-Analysis/](https://github.com/lewis-tang/Cluster-Analysis/)

References
[1] C. D. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval. Cambridge, U.K.: Cambridge University Press, 2008. [2] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods in Natural Language Processing and 9th Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, 2019, pp. 3982–3992. [3] K. Song, X. Tan, T. Qin, J. Lu, and T.-Y. Liu, "MPNet: Masked and permuted pre-training for language understanding," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 16857–16867. [4] B. Rogerson, "Taylor Swift reveals the 3 lyric 'genre categories' she uses when she's writing songs," MusicRadar, Sep. 21, 2022. [Online]. Available: www.musicradar.com/news/taylor-swift-songwriting-secret. [Accessed: Jun. 8, 2026]. [5] I. Johari, "Taylor Swift All Lyrics (64 Albums) [with TTPD]," Kaggle, 2024. [Online]. Available: <https://www.kaggle.com/datasets/ishikajohari/taylor-swift-all-lyrics-30-albums>. [Accessed: Jun. 8, 2026].

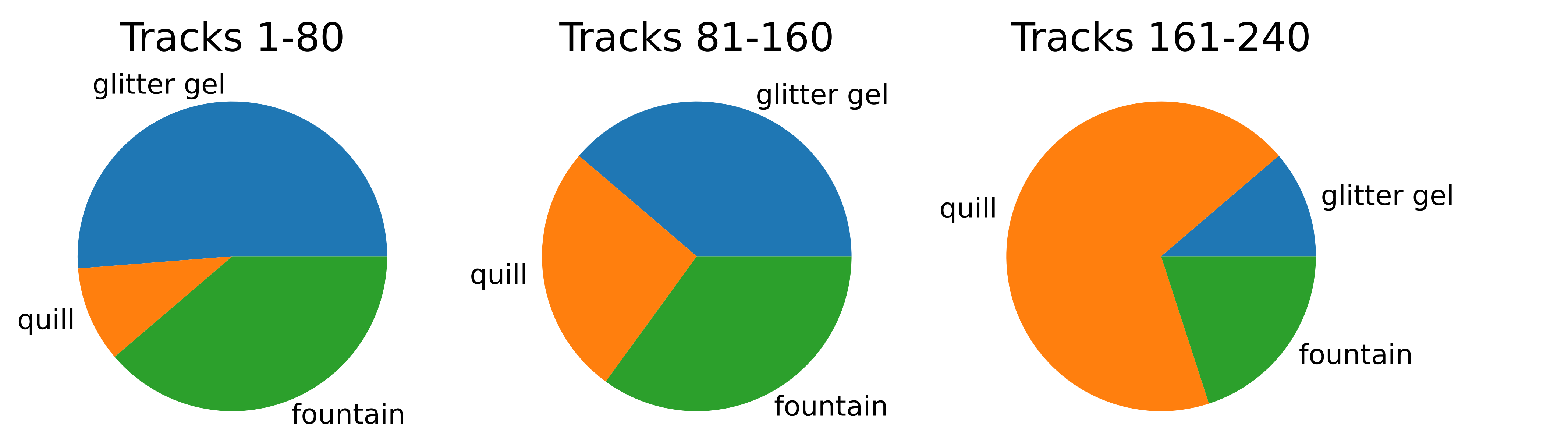


Figure 1: Sentence Transformer pipeline. Cluster names were assigned using context. Ordered chronologically by album.